JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON THURSDAY 22nd JANUARY 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. Let $\overrightarrow{AB} = 2\hat{i} + 4\hat{j} - 5k$ and $\overrightarrow{AD} = \hat{i} + 2\hat{j} + \lambda k, \lambda \in \mathbb{R}$. Let the projection of the vector $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ on the diagonal $\overrightarrow{AC}$ of the parallelogram ABCD be of length one unit. If α, β, where $\alpha > \beta$, be the roots of the equation $\lambda^2x^2 - 6\lambda x + 5 = 0$, then $2\alpha - \beta$ is equal to

(1) 1 (2) 4

(3) 3 (4) 6

2. Let the relation R on the set M = {1, 2, 3,.....16} be given by

R = {(x,y) : 4y = 5x – 3, x, y ∈ M}.

Then the minimum number of elements required to be added in R, in order to make the relation symmetric, is equal to

(1) 1 (2) 2

(3) 4 (4) 3

3. Let the line x = –1 divide the area of the region $\{(x,y) : 1 + x^2 \le y \le 3 - x\}$ in the ratio m : n, gcd (m, n) = 1. Then m + n is equal to

(1) 25 (2) 28

(3) 26 (4) 27

4. Two distinct numbers a and b are selected at random from 1, 2, 3,....., 50. The probability, that their product ab is divisible by 3, is

(1) $\frac{561}{1225}$ (2) $\frac{664}{1225}$

(3) $\frac{272}{1225}$ (4) $\frac{8}{25}$

5. Let $f(x) = x^{2025} - x^{2000}$, $x \in [0, 1]$ and the minimum value of the function f(x) in the interval [0, 1] be $(80)^{80}(n)^{-81}$. Then n is equal to

(1) –81 (2) –40

(3) –41 (4) –80

6. Let P(α, β, γ) be the point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance $4\sqrt{14}$ from the point (1, –1, 0) and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1} = \frac{y-\beta}{2} = \frac{z-\gamma}{3}$ and $\frac{x+5}{2} = \frac{y-10}{1} = \frac{z-3}{1}$, is equal to

(1) $7\sqrt{\frac{5}{4}}$ (2) $4\sqrt{\frac{7}{5}}$

(3) $4\sqrt{\frac{5}{7}}$ (4) $2\sqrt{\frac{7}{4}}$

7. If a random variable x has the probability distribution

x	0	1	2	3	4	5	6	7
p(x)	0	2k	k	3k	$2k^2$	2k	k^2+k	$7k^2$

then P(3 < x ≤ 6) is equal to

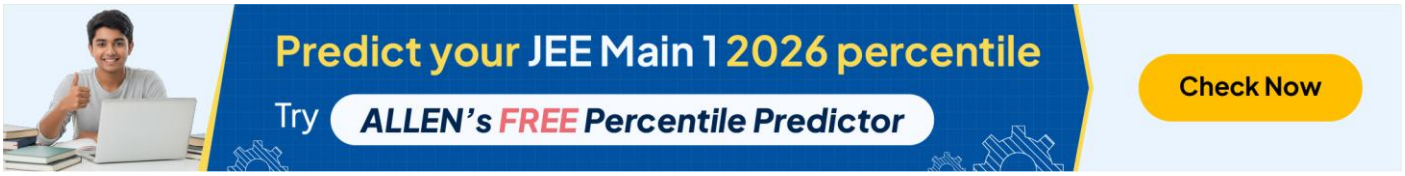
(1) 0.34 (2) 0.22

(3) 0.64 (4) 0.33

8. The number of distinct real solutions of the equation $x|x + 4| + 3|x + 2| + 10 = 0$ is

(1) 3 (2) 1

(3) 0 (4) 2



9. Let $f : [1, \infty) \to \mathbb{R}$ be a differentiable function, If $6\int_1^x f(t)dt = 3xf(x) + x^3 - 4$ for all x ≥ 1, then the value of $f(2) - f(3)$ is

(1) –4 (2) –3

(3) 4 (4) 3

10. If the line $\alpha x + 2y = 1$, where $\alpha \in \mathbb{R}$, does not meet the hyperbola $x^2 - 9y^2 = 9$, then a possible value of α is :

(1) 0.6 (2) 0.8

(3) 0.5 (4) 0.7

11. If the image of the point P (1, 2 , a) in the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{7-z}{2}$ is Q(5, b, c), then $a^2 + b^2 + c^2$ is equal to

(1) 293 (2) 264

(3) 298 (4) 283

12. Let the set of all values of r, for which the circles $(x + 1)^2 + (y + 4)^2 = r^2$ and $x^2 + y^2 - 4x - 2y - 4 = 0$ intersect at two distinct points be the interval (α, β). Then αβ is equal to

(1) 25 (2) 20

(3) 21 (4) 24

13. If $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, then the determinant of the matrix $(A^{2025} - 3A^{2024} + A^{2023})$ is

(1) 28 (2) 12

(3) 24 (4) 16

14. If the domain of the function $f(x) = \sin^{-1}\left(\frac{5-x}{3+2x}\right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$, then $6(\alpha + \beta + \gamma + \delta)$ is equal to

(1) 70 (2) 66

(3) 67 (4) 68

15. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{[x]+4}\right)dx$, where [•] denotes the greatest integer function, is

(1) $\frac{1}{60}(21\pi - 1)$

(2) $\frac{1}{60}(\pi - 7)$

(3) $\frac{7}{60}(3\pi - 1)$

(4) $\frac{7}{60}(\pi - 3)$

16. The coefficient of x^{48} in $(1 + x) + 2(1 + x)^2 + 3(1 + x)^3 + ... + 100(1 + x)^{100}$ is equal to :

(1) $100.\ {}^{100}C_{49} - {}^{100}C_{50}$ (2) ${}^{100}C_{50} + {}^{101}C_{49}$

(3) $100.\ {}^{100}C_{49} - {}^{100}C_{48}$ (4) $100.\ {}^{101}C_{49} - {}^{100}C_{50}$

17. If the chord joining the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1x_2 - y_1y_2$ is equal to

(1) 288 (2) 280

(3) 284 (4) 292

18. The number of solutions of $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$ is equal to

(1) 3 (2) 0

(3) 1 (4) 2

19. Let the solution curve of the differential equation $xdy - ydx = \sqrt{x^2 + y^2}\,dx$, x > 0, y(1) = 0, be y = y(x). Then y(3) is equal to

(1) 4 (2) 6

(3) 1 (4) 2



20. If the sum of the first four terms of an A.P. is 6 and the sum of its first six terms is 4, then the sum of its first twelve terms is

(1) –20 (2) –24

(3) –26 (4) –22

SECTION-B

21. Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $\beta = \frac{-1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. If $(7 - 7\alpha + 9\beta)^{20} + (9 + 7\alpha - 7\beta)^{20} + (-7 + 9\alpha + 7\beta)^{20} + (14 + 7\alpha + 7\beta)^{20} = m^{10}$, then m is _____.

22. Let A be a 3 × 3 matrix such that $A + A^T = O$. If $A\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}3\\3\\2\end{bmatrix}, A^2\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}-3\\19\\-24\end{bmatrix}$ and $\det(\text{adj}(2\text{adj}(A + I))) = (2)^{\alpha}.(3)^{\beta}.(11)^{\gamma}$, α, β, γ are non-negative integers, then $\alpha + \beta + \gamma$ is equal to _____

23. If $\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = -\frac{p_1}{q_1}(\cot x)^{\frac{9}{2}} - \frac{p_2}{q_2}(\cot x)^{\frac{5}{2}} - \frac{p_3}{q_3}(\cot x)^{\frac{1}{2}} + \frac{p_4}{q_4}(\cot x)^{\frac{-3}{2}} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4}$ is equal to _____.

24. If $\frac{\cos^2 48° - \sin^2 12°}{\sin^2 24° - \sin^2 6°} = \frac{\alpha + \beta\sqrt{5}}{2}$, where $\alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to _____.

25. Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC and four points p_{10}, p_{11}, p_{12}, p_{13} on the side AC. None of these points is a vertex of the triangle ABC. Then the total number of pentagons, that can be formed by taking all the vertices from the points p_1, p_2, ... p_{13}, is _____.



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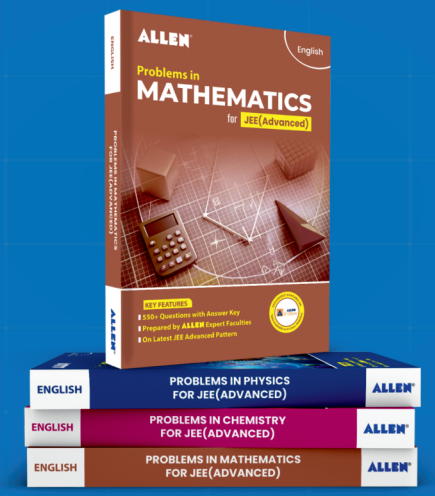


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